

ShopEasy Marketing Analytics — End-to-End Data Analyst Project

An end-to-end marketing analytics project for a simulated online retail business, **ShopEasy**, built to diagnose declining customer engagement and conversion rates despite increased marketing spend. The project moves through the full analytics lifecycle: business problem framing → SQL data extraction & cleaning → Python-based sentiment enrichment → Power BI data modeling & dashboarding → stakeholder-ready PowerPoint storytelling.

Business Problem

ShopEasy, an online retail business, was facing:

- **Reduced customer engagement** with site content and marketing materials
- **Decreased conversion rates** despite growing traffic
- **Rising marketing expenses** without proportional ROI
- **Unstructured customer feedback** (reviews, social comments) that hadn't been systematically analyzed

Stakeholder requests (Marketing Manager, Customer Experience Manager) were translated into a defined scope with KPIs — **Conversion Rate, Customer Engagement Rate, Average Order Value, Customer Feedback Score** — and three measurable goals:

1. Increase conversion rates
 2. Enhance customer engagement
 3. Improve customer feedback scores
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Project Workflow

1. Business Case & Requirements Gathering

- Reviewed stakeholder briefs to extract pain points, KPIs, and success criteria
- Mapped data sources (customer journey, engagement data, reviews, geography, products) to fact/dimension roles ahead of modeling

2. Data Extraction & Cleaning — SQL Server

- Wrote transformation queries per source table rather than cleaning downstream, including:
 - `CASE` logic to bucket product price into Low/Medium/High categories
 - `LEFT JOIN` to combine `customers` and `geography` into a single dimension table
 - `REPLACE` / `UPPER` to standardize inconsistent text (double spaces in reviews, mixed-case content types)
 - Splitting a combined `views/clicks` column into separate numeric fields
 - **Duplicate detection and removal** using `ROW_NUMBER() OVER (PARTITION BY ...)` inside a CTE

- **Missing value imputation** using a correlated subquery that filled null `duration` values with the average duration for that visit date

3. Sentiment Enrichment — Python

- Connected to the SQL Server database directly from Python (pandas + SQL connector)
- Applied sentiment scoring (NLTK) to customer review text, producing a **sentiment score (-1 to 1)**
- Built a custom function combining sentiment score **with** the numeric star rating to flag mismatches (e.g., a 3-star review with strongly negative text → "mixed negative"), surfacing signal that rating-only analysis would miss
- Bucketed scores into sentiment categories and exported the enriched dataset for BI ingestion

4. Data Modeling & Dashboarding — Power BI

- Built a **star schema**: fact tables (Customer Journey, Customer Reviews w/ sentiment, Engagement Data) and dimension tables (Customers+Geography, Products, Calendar)
- Created a dedicated measures table with DAX calculations including a **conversion rate measure using variables** (`VAR total_visitors`, `VAR total_purchases`) with divide-by-zero handling
- Delivered a 4-page interactive report:
 - **Overview** — high-level KPIs and trends across conversion, social engagement, and reviews
 - **Conversion Details** — funnel visualization and conversion rate by product/month
 - **Social Media Details** — engagement by content type, conditional-formatted matrix
 - **Customer Review Details** — rating distribution, sentiment trends over time, scatter plot clustering by sentiment category

5. Insight Presentation — PowerPoint

- Translated dashboard findings into a stakeholder-facing narrative deck, mapped back to the original business goals
- Used specific, quantified observations rather than generic statements, e.g.:
 - Conversion rate dipped to 5% in October before rebounding to 10.2% in December
 - Clicks/likes held at ~15-37% engagement even as raw views declined
 - Average rating held steady at 3.7, below the 4.0 target — with mixed-sentiment reviews identified as the improvement lever
- Closed with a goals → insight → recommended action structure tying every recommendation back to a KPI defined in the original business case

Tech Stack

Layer	Tools
Data Storage & Transformation	SQL Server (T-SQL)
Enrichment / NLP	Python (pandas, NLTK)
Modeling & Visualization	Power BI (Power Query, DAX)
Presentation	PowerPoint

Key Skills Demonstrated

- Translating stakeholder requirements into measurable KPIs
- SQL: joins, CASE logic, window functions, CTEs, correlated subqueries, data cleaning
- Python: database connectivity, NLP-based sentiment analysis, custom scoring logic
- Data modeling: star schema design, fact/dimension separation
- DAX: measures, variables, conditional logic
- Dashboard design: multi-page reports, conditional formatting, funnel/scatter visualizations
- Data storytelling: converting technical findings into stakeholder-appropriate narrative

Repository Structure

```
├─ sql/           # Extraction & cleaning queries per table
├─ python/        # Sentiment analysis script + output CSV
├─ powerbi/       # .pbix file, data model, DAX measures
├─ presentation/  # Stakeholder PowerPoint deck
└─ docs/          # Business case brief, KPI definitions, goals
```

How to Reproduce

1. Restore the provided SQL Server backup and run the scripts in `/sql` to generate cleaned tables
2. Run `/python/sentiment_analysis.py` to generate the enriched reviews CSV
3. Open the Power BI file in `/powerbi`, refresh data sources, and review the model/measures
4. Reference `/presentation` for the summarized business narrative

Notes

This project was completed as a structured learning exercise based on a public data analyst portfolio case study, then used to practice the full SQL → Python → Power BI → presentation

workflow end-to-end. All queries, transformations, and dashboard design decisions were implemented and understood individually rather than copied without review.